

PROJECT ABSTRACT[Form-1] (YEAR-2026-27)

PROJECT ID: SKIT/CSE/2023-2027/26 BRANCH: Computer Science & Engineering SECTION: B

TITLE OF PROJECT: SecureCred – Blockchain Based Digital Certification System

SDG MAPPING: SDG 4 (Write SDG Title) Quality Education

PROJECT TRACK: (Tick the appropriate one / ones)

R&D INNOVATION (Selected)	CONSULTANCY PROJECTS	DEPARTMENT AS CLIENT PROJECTS	CLIENT PROJECT	HARDWARE-BASED PROJECTS
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PROBLEM STATEMENT:

Existing academic and vocational certification systems rely on paper certificates or plain PDF/centralized records, which are vulnerable to forgery and tampering and offer no cryptographic proof of authenticity. Verifying a certificate typically requires manually contacting the issuing institute, which is slow and unreliable, and leaves no audit trail for issuance or revocation. Learners can also permanently lose access to their credentials if an institute shuts down or misplaces its records.

SecureCred (Blockchain Based Digital Certification System) securely issues and verifies vocational credentials by cryptographically hashing (SHA-256) each certificate, storing it via decentralized IPFS/Pinata storage, and anchoring the hash on the Polygon blockchain — enabling instant verification while reducing manual effort, cost, fraud, and the risk of permanent credential loss.

TECHNOLOGY STACK:

NAME OF TOOL / TECHNOLOGY	FRONTEND/BACKEND	SOFTWARE / HARDWARE	PURPOSE OF USE
React.js	Frontend	Software	Build responsive user interface and dashboards
Tailwind CSS	Frontend	Software	Utility-first styling for responsive UI design
Figma	Frontend	Software	UI/UX prototyping and interface design
Node.js	Backend	Software	Server-side JavaScript runtime environment
Express.js	Backend	Software	REST API development and backend routing
Convex	Backend	Software	Backend application and data services
PostgreSQL	Database	Software	Store off-chain certificate metadata
IPFS	Storage	Software	Decentralized storage for certificate files
Pinata	Storage	Software	IPFS pinning service for reliable file access
SHA-256	Security	Software	Generate cryptographic hash of each certificate
Solidity	Blockchain	Software	Develop smart contracts for

			issue/verify/revoke logic
Polygon	Blockchain	Software	Anchor certificate hash & CID on-chain (low-cost network)
Ganache	Blockchain	Software	Local blockchain environment for smart contract testing
Docker	Deployment	Software	Containerize application for deployment
GitHub	Deployment	Software	Version control and team collaboration

PROPOSED PROJECT SPRINTS (add more row if required):

S.N o.	SPRINT NAME	PROPOSED USER STORY OF RESPECTIVE SPRINT
1	System Architecture & Setup	As a project team, we want to design the system architecture, set up the GitHub repositories, backend/frontend project scaffolding, and the PostgreSQL database schema so that the technical foundation for parallel development is ready. (Milestone: 20% by 22/08/2026)
2	Core Module Development	As a project team, we want to develop the core smart contract, backend REST APIs and frontend UI/dashboard components in parallel so that the primary certificate issuance modules are functional. (Milestone: 40% by 03/10/2026)
3	Feature Integration & Verification	As a project team, we want to implement certificate verification and revocation logic, authentication, IPFS/Pinata storage integration and the verifier portal so that end-to-end issuance and verification works. (Milestone: 50% by 31/10/2026)
4	Backend-Blockchain-Frontend Integration	As a project team, we want to integrate the backend, blockchain and frontend layers and deploy smart contracts to a public testnet so that the complete workflow operates across all modules. (Milestone: 70% by 02/01/2027)
5	Security Hardening & Testing	As a project team, we want to conduct smart contract security audits, API security hardening, and end-to-end integration testing so that the system is secure and reliable before release. (Milestone: 90% by 06/02/2027)
6	Final Deployment & Release	As a project team, we want to complete regression testing, finalize production deployment on mainnet/cloud infrastructure, and release the system so that SecureCred is ready for use. (Milestone: 100% by 06/03/2027)

TEAM MEMBER DETAILS:

STUDENT NAME	MOBILE No.	EXPERTISE AREA	ROLE IN PROJECT
Kartik Marwal	8386847113	Blockchain, Backend	Team Lead, Blockchain & System Architecture
Jaideep Singh Chauhan	8949567353	Backend, Database	Backend API, PostgreSQL, Authentication
Keshav Bheel	7850947634	Frontend	UI Development, React, Dashboard Design
Kavish Mangal	8435097127	Testing	Testing, Documentation, Security

Verified & Approved By:

(MENTOR'S NAME & SIGNATURE)

Project Mentor

ROLES & RESPONSIBILITY OF TEAM MEMBERS [Form – 2]

PROJECT ID: SKIT/CSE/2023-2027/26 BRANCH: Computer Science & Engineering SECTION: B

TITLE OF PROJECT: SecureCred – Blockchain Based Digital Certification System

TEAM LEAD : Kartik Marwal		SPRINT(s) NAME : System Architecture & Blockchain Design, Smart Contract Development - Issuance & Storage, Verification & Revocation Contract Logic, Polygon Testnet Deployment & Gas Optimization, Smart Contract Security Audit & Hardening, Mainnet Deployment & Final Blockchain Integration	
USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
System Architecture & Blockchain Design	10/08/2026	22/08/2026	Design the system architecture, blockchain workflow and certificate lifecycle connecting frontend, backend, IPFS and Polygon. (20% milestone)
Smart Contract Development - Issuance & Storage	23/08/2026	03/10/2026	Develop the Solidity smart contract for certificate issuance and hash storage; deploy and test on Ganache. (40% milestone)
Verification & Revocation Contract Logic	04/10/2026	31/10/2026	Implement on-chain certificate hash verification and issuer-controlled revocation logic; test on Ganache. (50% milestone)
Polygon Testnet Deployment & Gas Optimization	01/11/2026	02/01/2027	Deploy smart contracts to the Polygon Amoy testnet, optimize gas usage, and integrate contract events with backend listeners. (70% milestone)
Smart Contract Security Audit & Hardening	03/01/2027	06/02/2027	Conduct a security audit of the smart contracts (reentrancy, access-control checks) and patch identified vulnerabilities. (90% milestone)
Mainnet Deployment & Final Blockchain Integration	07/02/2027	15/03/2027	Finalize contract deployment on the Polygon production network and complete end-to-end blockchain integration testing. (100% milestone)

TEAM MEMBER : Jaideep Singh Chauhan		SPRINT(s) NAME : Backend Setup & DB Schema Design, Certificate REST API Development, Authentication Module & IPFS/Pinata Integration, Backend-Blockchain Integration, API Security Hardening & Rate Limiting, Backend Deployment & Production Readiness	
USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
Backend Setup & DB Schema Design	10/08/2026	22/08/2026	Set up the Node.js/Express backend project structure and design the PostgreSQL database schema for certificate metadata storage. (20% milestone)
Certificate REST API Development	23/08/2026	03/10/2026	Develop REST API endpoints for certificate creation, retrieval and metadata management. (40% milestone)
Authentication Module & IPFS/Pinata Integration	04/10/2026	31/10/2026	Implement JWT-based authentication with role-based access control, and integrate the Pinata SDK to upload certificate files to IPFS. (50% milestone)
Backend-Blockchain Integration	01/11/2026	02/01/2027	Connect backend APIs with smart contract functions for issuance, verification and revocation, including transaction status tracking. (70% milestone)
API Security Hardening & Rate Limiting	03/01/2027	06/02/2027	Implement input validation, rate limiting, and fix vulnerabilities identified during security testing. (90% milestone)

Backend Deployment & Production Readiness	07/02/2027	15/03/2027	Containerize the backend with Docker, configure the production database, and finalize deployment scripts. (100% milestone)
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TEAM MEMBER : Keshav Bheel		SPRINT(s) NAME : UI Component Library Development , Institution & Student Dashboard Development, Verifier Portal - QR/ID Verification UI, Frontend-Backend & Blockchain API Integration, UI/UX Refinement & Bug Fixing, Frontend Production Deployment	
USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
UI Component Library Development	10/08/2026	22/08/2026	Build reusable React.js UI components and convert Figma layouts into functional code using Tailwind CSS. (20% milestone)
Institution & Student Dashboard Development	23/08/2026	03/10/2026	Develop the institution dashboard for certificate issuance/management and the student dashboard for viewing issued certificates. (40% milestone)
Verifier Portal - QR/ID Verification UI	04/10/2026	31/10/2026	Build the verifier portal with QR-code scanning and ID-based certificate verification flow. (50% milestone)
Frontend-Backend & Blockchain API Integration	01/11/2026	02/01/2027	Integrate frontend dashboards with backend REST APIs and blockchain transaction status for live certificate data. (70% milestone)
UI/UX Refinement & Bug Fixing	03/01/2027	06/02/2027	Refine UI/UX based on testing feedback and fix bugs identified during integration testing. (90% milestone)
Frontend Production Deployment	07/02/2027	15/03/2027	Containerize and deploy the frontend application to production using Docker. (100% milestone)

TEAM MEMBER : Kavish Mangal		SPRINT(s) NAME : Test Automation Framework Setup, Smart Contract Unit Test Development, Backend API Test Automation, Security Testing - Tamper & Access Control, End-to-End Integration Testing, Regression Testing & Release Validation	
USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
Test Automation Framework Setup	10/08/2026	22/08/2026	Set up the Hardhat/Chai test framework and Ganache local blockchain environment for automated smart contract testing. (20% milestone)
Smart Contract Unit Test Development	23/08/2026	03/10/2026	Write automated unit test scripts covering certificate issuance, verification and revocation contract functions. (40% milestone)
Backend API Test Automation	04/10/2026	31/10/2026	Develop automated test scripts (Postman/Jest) for backend REST API endpoints. (50% milestone)
Security Testing - Tamper & Access Control	01/11/2026	02/01/2027	Perform security testing for certificate hash-tampering detection and role-based access control vulnerabilities. (70% milestone)
End-to-End Integration Testing	03/01/2027	06/02/2027	Conduct integration testing across frontend, backend and blockchain components to validate the complete certificate issuance and verification flow. (90% milestone)
Regression Testing & Release Validation	07/02/2027	15/03/2027	Fix identified bugs, execute regression testing, and validate system readiness for final release deployment. (100% milestone)

Verified & Approved By:

(MENTOR'SNAME&SIGNATURE)

Project Mentor